

2(a)(i)	Man is moving at maximum speed when resultant force acting on him is zero, hence $mg = kx$ where x is the extension of the rope at that point. $x = \frac{mg}{k} = \frac{(80.0)(9.81)}{120}$ $= 6.54 \text{ m}$	[1] sub
(a)(ii)	Applying Principle of Conservation of Energy to when the man is at maximum speed, Loss of GPE = Gain in KE + Gain in Elastic PE $mg(l + x) = \frac{1}{2}mv_{\text{max}}^2 + \frac{1}{2}kx^2$ $(80.0)(9.81)(25.0 + 6.54) = \frac{1}{2}(80.0)v_{\text{max}}^2 + \frac{1}{2}(120)(6.54)^2$ $v_{\text{max}} = 23.6 \text{ m s}^{-1}$	[1] sub [1] ans
(a)(iii)	$\text{Maximum KE} = \frac{1}{2}(80.0)v_{\text{max}}^2 = \frac{1}{2}(80.0)(23.6)^2 = 2.23 \times 10^4 \text{ J}$	[1]

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Suggested Solutions with Markers' Comments

Qn	Suggested solution	Remarks
(b)(i)	Applying Principle of Conservation of Energy at the lowest point, Loss of GPE = Gain in Elastic PE $mg(l + x_{\max}) = \frac{1}{2}kx_{\max}^2$ $(80.0)(9.81)(25.0 + x_{\max}) = \frac{1}{2}(120)x_{\max}^2$ $x_{\max} = 25.8 \text{ m}$	[1] sub [1] ans
(b)(ii)	$\text{GPE} = mg(l + x_{\max}) = (80.0)((9.81)(25.0 + 25.8) = 3.99 \times 10^4 \text{ J}$	[1]
MC		
(c) (i) (ii) (iii)		[1] G with values [1] E with values [1] K with values
3(a)(i)		